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Project: TraceRoute

INTRODUCTION:

This project uses ICMP to create a traceroute tool that shows the path a packet takes to reach a remote host. The program sends out echo requests and slowly increases the TTL values to force each router along the way to send back a Time Exceeded (Type 11) message. This continues until the packet finally hits the destination, which sends back an Echo Reply (Type 0). To make sure the data is accurate, the app includes a validation layer that checks the sequence number, packet identifier, and raw data for every response. It also uses a lookup structure to turn ICMP error codes into human-readable messages.

INSTRUCTIONS:

Programming Language: Python 3

Download File: icmp_helper_library.py

Firewall Rule: Add a new rule to your operating system to allow ICMP messages.

> Windows: Go to “Windows Defender Firewall - Advanced Settings” and click “Inbound Rules”. Click “New Rule” > “Custom” > “All Programs” > and select “ICMPv4” for protocol type. Press next until the rule is created.
>
> MacOS/Linux: For local firewalls such as UFW, run “sudo ufw allow icmp”. You may also need to temporarily disable third party antivirus firewalls that block raw sockets.

Execution: Run the icmp_helper_library.py script from your terminal using administrator privileges. Go to the main function and uncomment testing routes as needed. Feel free to enter your own route for testing!

CITATIONS:

The ICMP parameters and error code lookup logic was adapted from the following official registry:

> IANA (2026) Internet Control Message Protocol (ICMP) Parameters [Online Registry].
> https://www.iana.org/assignments/icmp-parameters/icmp-parameters.xhtml

The socket logic and traceroute implementation concepts were adapted from the following technical guide:

> GeeksforGeeks (2023) Traceroute implementation on Python [Technical Guide].
> https://www.geeksforgeeks.org/linux-unix/traceroute-implementation-on-python/

SAMPLE PING FUNCTIONALITY:

Host #1: gaia.cs.umass.edu (University of Massachusetts - Networking Server)

Function Call: icmpHelperPing.sendPing("gaia.cs.umass.edu")

```
EXP: ID=4792, Seq=0, Data=ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz
ACT: ID=4792, Seq=0, Data=ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz
 TTL=255 RTT=59 ms Type=0 Code=0 (ECHO RESPONSE - Destination Reached) from 128.119.245.12

EXP: ID=4792, Seq=1, Data=ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz
ACT: ID=4792, Seq=1, Data=ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz
 TTL=255 RTT=58 ms Type=0 Code=0 (ECHO RESPONSE - Destination Reached) from 128.119.245.12

EXP: ID=4792, Seq=2, Data=ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz
ACT: ID=4792, Seq=2, Data=ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz
 TTL=255 RTT=59 ms Type=0 Code=0 (ECHO RESPONSE - Destination Reached) from 128.119.245.12

--- Ping statistics for gaia.cs.umass.edu ---
 Packets: Sent=3, Received=3, Lost=0 (0.0% loss)
 Min=58ms, Max=59ms, Avg=58.666666666666664ms
```

Host #2: www.google.com (Google - US Server)

Function Call: icmpHelperPing.sendPing("www.google.com")

```
EXP: ID=8060, Seq=0, Data=ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz
ACT: ID=8060, Seq=0, Data=ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz
 TTL=255 RTT=35 ms Type=0 Code=0 (ECHO RESPONSE - Destination Reached) from 142.251.41.68

EXP: ID=8060, Seq=1, Data=ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz
ACT: ID=8060, Seq=1, Data=ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz
 TTL=255 RTT=34 ms Type=0 Code=0 (ECHO RESPONSE - Destination Reached) from 142.251.41.68

EXP: ID=8060, Seq=2, Data=ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz
ACT: ID=8060, Seq=2, Data=ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz
 TTL=255 RTT=36 ms Type=0 Code=0 (ECHO RESPONSE - Destination Reached) from 142.251.41.68

--- Ping statistics for www.google.com ---
 Packets: Sent=3, Received=3, Lost=0 (0.0% loss)
 Min=34ms, Max=36ms, Avg=35.0ms
```

Host #3: baidu.com (China - National Search Engine)

Function Call: icmpHelperPing.sendPing("*baidu.com*")

```
EXP: ID=10836, Seq=0, Data=ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz
ACT: ID=10836, Seq=0, Data=ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz
 TTL=255 RTT=250 ms Type=0 Code=0 (ECHO RESPONSE - Destination Reached) from 111.63.65.103

EXP: ID=10836, Seq=1, Data=ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz
ACT: ID=10836, Seq=1, Data=ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz
 TTL=255 RTT=250 ms Type=0 Code=0 (ECHO RESPONSE - Destination Reached) from 111.63.65.103

EXP: ID=10836, Seq=2, Data=ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz
ACT: ID=10836, Seq=2, Data=ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz
 TTL=255 RTT=250 ms Type=0 Code=0 (ECHO RESPONSE - Destination Reached) from 111.63.65.103

--- Ping statistics for baidu.com ---
 Packets: Sent=3, Received=3, Lost=0 (0.0% loss)
 Min=250ms, Max=250ms, Avg=250.0ms
```

Host #4: eng.taiwan.net.tw (Taiwan - National Tourism Server)

Function Call: icmpHelperPing.sendPing("eng.taiwan.net.tw")

```
EXP: ID=62164, Seq=0, Data=ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz
ACT: ID=62164, Seq=0, Data=ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz
 TTL=255 RTT=175 ms Type=0 Code=0 (ECHO RESPONSE - Destination Reached) from 211.79.207.131

EXP: ID=62164, Seq=1, Data=ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz
ACT: ID=62164, Seq=1, Data=ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz
 TTL=255 RTT=177 ms Type=0 Code=0 (ECHO RESPONSE - Destination Reached) from 211.79.207.131

EXP: ID=62164, Seq=2, Data=ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz
ACT: ID=62164, Seq=2, Data=ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz
 TTL=255 RTT=174 ms Type=0 Code=0 (ECHO RESPONSE - Destination Reached) from 211.79.207.131

--- Ping statistics for eng.taiwan.net.tw ---
 Packets: Sent=3, Received=3, Lost=0 (0.0% loss)
 Min=174ms, Max=177ms, Avg=175.33333333333334ms
```

SAMPLE TRACEROUTE FUNCTIONALITY:

Host #1: gaia.cs.umass.edu (University of Massachusetts - Networking Server)

Function Call: icmpHelperPing.traceRoute("gaia.cs.umass.edu")

ICMP Type:11 and 0 Code:0

```
Tracing route to (gaia.cs.umass.edu) 128.119.245.12
  TTL=1    RTT=4 ms    Type=11 (TTL Exceeded)    Code=0    192.168.4.1
  TTL=2    RTT=6 ms    Type=11 (TTL Exceeded)    Code=0    170.203.104.2
  TTL=3    RTT=7 ms    Type=11 (TTL Exceeded)    Code=0    38.122.156.167
  *        *        *        *        *    Request timed out.
  *        *        *        *        *    Request timed out (By Exception).
  TTL=5    RTT=6 ms    Type=11 (TTL Exceeded)    Code=0    154.54.47.25
  TTL=6    RTT=14 ms    Type=11 (TTL Exceeded)    Code=0    154.54.170.130
  TTL=7    RTT=23 ms    Type=11 (TTL Exceeded)    Code=0    154.54.40.109
  TTL=8    RTT=37 ms    Type=11 (TTL Exceeded)    Code=0    154.54.172.169
  TTL=9    RTT=40 ms    Type=11 (TTL Exceeded)    Code=0    154.54.30.122
  TTL=10    RTT=47 ms    Type=11 (TTL Exceeded)    Code=0    154.54.46.33
  TTL=11    RTT=49 ms    Type=11 (TTL Exceeded)    Code=0    154.54.170.2
  TTL=12    RTT=48 ms    Type=11 (TTL Exceeded)    Code=0    154.54.160.70
  TTL=13    RTT=50 ms    Type=11 (TTL Exceeded)    Code=0    38.32.33.83
  TTL=14    RTT=54 ms    Type=11 (TTL Exceeded)    Code=0    69.16.0.9
  TTL=15    RTT=58 ms    Type=11 (TTL Exceeded)    Code=0    69.16.3.0
  TTL=16    RTT=60 ms    Type=11 (TTL Exceeded)    Code=0    192.80.83.105
  TTL=17    RTT=57 ms    Type=11 (TTL Exceeded)    Code=0    128.119.0.10
  TTL=18    RTT=55 ms    Type=11 (TTL Exceeded)    Code=0    128.119.77.235
  *        *        *        *        *    Request timed out.
  *        *        *        *        *    Request timed out (By Exception).
  TTL=20    RTT=56 ms    Type=11 (TTL Exceeded)    Code=0    128.119.3.32
  *        *        *        *        *    Request timed out.
  *        *        *        *        *    Request timed out (By Exception).
EXP: ID=6984, Seq=22, Data=ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz
ACT: ID=6984, Seq=22, Data=ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz
 TTL=22 RTT=57 ms Type=0 Code=0 (ECHO RESPONSE - Destination Reached) from 128.119.245.12

Destination gaia.cs.umass.edu reached.
```

Host #2: 122.56.99.243 (New Zealand - Telecom Company)

Function Call: icmpHelperPing.traceRoute("122.56.99.243") ***<u>Network Unreachable</u>***

ICMP Type:3 Code:0

```
Tracing route to (122.56.99.243) 122.56.99.243
  TTL=1    RTT=2 ms    Type=11 (TTL Exceeded)    Code=0    192.168.4.1
  TTL=2    RTT=3 ms    Type=11 (TTL Exceeded)    Code=0    170.203.104.2
  TTL=3    RTT=3 ms    Type=3 (Network Unreachable)    Code=0    213.248.67.118
```

Host #3: 168.176.239.10 (Columbia - Universidad Nacional de Colombia)

Function Call: icmpHelperPing.traceRoute("168.176.239.10") ***<u>Likely Hit University's Firewall</u>***

ICMP Type:11 and 0 Code:0

```
Tracing route to (168.176.239.10) 168.176.239.10
  TTL=1    RTT=5 ms    Type=11 (TTL Exceeded)    Code=0    192.168.4.1
  TTL=2    RTT=4 ms    Type=11 (TTL Exceeded)    Code=0    170.203.104.2
  *        *        *        *        *    Request timed out.
  *        *        *        *        *    Request timed out (By Exception).
  TTL=4    RTT=7 ms    Type=11 (TTL Exceeded)    Code=0    149.112.113.2
  TTL=5    RTT=6 ms    Type=11 (TTL Exceeded)    Code=0    184.105.64.245
  *        *        *        *        *    Request timed out.
  *        *        *        *        *    Request timed out (By Exception).
  TTL=7    RTT=31 ms    Type=11 (TTL Exceeded)    Code=0    198.32.242.254
  TTL=8    RTT=31 ms    Type=11 (TTL Exceeded)    Code=0    200.16.69.65
  TTL=9    RTT=65 ms    Type=11 (TTL Exceeded)    Code=0    200.16.69.61
  TTL=10    RTT=67 ms    Type=11 (TTL Exceeded)    Code=0    138.0.42.135
  TTL=11    RTT=95 ms    Type=11 (TTL Exceeded)    Code=0    186.33.56.81
  TTL=12    RTT=90 ms    Type=11 (TTL Exceeded)    Code=0    186.33.56.82
  *        *        *        *        *    Request timed out.
  *        *        *        *        *    Request timed out (By Exception).
  *        *        *        *        *    Request timed out.
  *        *        *        *        *    Request timed out (By Exception).
  *        *        *        *        *    Request timed out.
  *        *        *        *        *    Request timed out (By Exception).
  *        *        *        *        *    Request timed out.
  *        *        *        *        *    Request timed out (By Exception).
  *        *        *        *        *    Request timed out.
  *        *        *        *        *    Request timed out (By Exception).
  *        *        *        *        *    Request timed out.
  *        *        *        *        *    Request timed out (By Exception).
  *        *        *        *        *    Request timed out.
```

Host #4: www.baidu.com (China - National Search Engine)

Function Call: icmpHelperPing.traceRoute("www.baidu.com")

ICMP Type:11 and 0 Code:0

```
Tracing route to (www.baidu.com) 103.235.46.115
  TTL=1    RTT=3 ms    Type=11 (TTL Exceeded)    Code=0    192.168.4.1
  TTL=2    RTT=4 ms    Type=11 (TTL Exceeded)    Code=0    170.203.104.2
  TTL=3    RTT=5 ms    Type=11 (TTL Exceeded)    Code=0    213.248.67.118
  TTL=4    RTT=10 ms    Type=11 (TTL Exceeded)    Code=0    62.115.123.106
  *        *        *        *        *    Request timed out.
  *        *        *        *        *    Request timed out (By Exception).
  TTL=6    RTT=13 ms    Type=11 (TTL Exceeded)    Code=0    62.115.139.130
  TTL=7    RTT=12 ms    Type=11 (TTL Exceeded)    Code=0    64.86.132.238
  *        *        *        *        *    Request timed out.
  *        *        *        *        *    Request timed out (By Exception).
  TTL=9    RTT=221 ms    Type=11 (TTL Exceeded)    Code=0    64.86.252.33
  *        *        *        *        *    Request timed out.
  *        *        *        *        *    Request timed out (By Exception).
  *        *        *        *        *    Request timed out.
  *        *        *        *        *    Request timed out (By Exception).
  *        *        *        *        *    Request timed out.
  *        *        *        *        *    Request timed out (By Exception).
  TTL=13    RTT=196 ms    Type=11 (TTL Exceeded)    Code=0    180.87.162.46
  *        *        *        *        *    Request timed out.
  *        *        *        *        *    Request timed out (By Exception).
  *        *        *        *        *    Request timed out.
  *        *        *        *        *    Request timed out (By Exception).
  *        *        *        *        *    Request timed out.
  *        *        *        *        *    Request timed out (By Exception).
EXP: ID=10464, Seq=17, Data=ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz
ACT: ID=10464, Seq=17, Data=ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz
 TTL=17 RTT=198 ms Type=0 Code=0 (ECHO RESPONSE - Destination Reached) from 103.235.46.115
```

COMMENTS:

Implementing traceroute showed exactly how the IP TTL field and ICMP error reporting works in the real world. By slowly bumping up the TTL, the program forces each router to send back a Type 11 (TTL Exceeded) message to map out the path.

One thing that really stood out was the massive jump in RTT when I tried pinging hosts in China or Taiwan. You can actually see the exact hop where the packet hits the undersea cables because the latency spikes from 20ms to over 200ms instantly. I spent some time trying to get a Type 3 Code 1 (Host Unreachable) on my local network, but it kept timing out. I believe either my router settings or firewall is in stealth mode, so it just drops the packet entirely. Getting that Type 3 Code 0 from the New Zealand host proved the IANA parsing works correctly.